

Optimisation-Based Manipulation Planning in Convex Decompositions of the Composite Configuration Space

Julia López Gómez and Steve Tonneau [CHECK: author list and order]
School of Informatics, University of Edinburgh, United Kingdom
[TODO: emails]

Abstract—Manipulation planning requires a robot to decide where to grasp, transport and release an object while moving through a high-dimensional configuration space shaped by collisions, joint limits and grasp constraints. These conditions make the feasible space narrow, disconnected and hard to describe. Sampling-based manipulation planners are probabilistically complete but converge slowly in the narrow, lower-dimensional sets where grasps and regrasps happen, while trajectory optimisation can exploit problem structure but is sensitive to initialisation and offers no completeness guarantees. We propose a manipulation planner that operates on a convex decomposition of the collision-free composite configuration space of robot and object. Grasp and placement spaces are represented as unions of convex polytopes and intersected with the decomposition, so that the classical manipulation graph becomes a graph of convex sets. A minimum-regrasp sequence of grasp and release configurations is selected by a mixed-integer quadratic program, or by heuristic search over the polytope graph, and connected by globally optimal transit and transfer trajectories computed with Graphs of Convex Sets. For high-dimensional arms, where grasping is a nonlinear constraint on the forward kinematics, we generate diverse feasible grasp configurations with NLP sampling and build grasp polytopes aligned with the null space of the active grasp constraints. The resulting planner is complete and optimal relative to the constructed decomposition. We evaluate it on bottle-cap opening tasks that require repeated regrasping, from 3- and 4-DOF systems up to the 10-dimensional composite space of a Franka Panda arm, hand and cap, and compare it with sampling-based baselines. [TODO: one sentence with the headline result once experiments are in.]

Index Terms—Manipulation planning, motion and path planning, convex optimisation, graphs of convex sets, constrained sampling.

I. INTRODUCTION

Manipulating an object involves more than moving a robot without collision. The robot must decide where to grasp the object, how to carry it, where to release it and, when its own kinematics prevent the task from being completed in one motion, where to regrasp it. Geometrically, this is a motion planning problem in the *composite* configuration space of robot and object [1], [2], where the object can only move while

it is grasped. Valid solutions therefore alternate between *transit* paths, in which the robot moves alone while the object rests at a stable placement, and *transfer* paths, in which the robot moves while holding the object at a fixed grasp. Switching between the two is only possible on the intersection of the grasp and placement spaces, a set of measure zero in the composite space. Together with collision avoidance and joint limits, this makes the feasible space narrow, disconnected and highly nonlinear.

The two dominant families of planners handle this structure in complementary ways. Sampling-based planners build roadmaps or trees over sampled configurations [3], [4], and their extension to manipulation through the manipulation graph [2] and multi-modal planning [5] gives probabilistic completeness and, for some variants, asymptotic optimality [6], [7]. In practice, however, they degrade with dimension and in narrow passages, and require dedicated samplers to hit the lower-dimensional sets where grasps happen. Trajectory optimisation [8], [9] instead exploits gradients and optimises task costs directly, but collision and grasp constraints are non-convex, so solutions are local, depend on initialisation, and failure gives no information about whether a solution exists.

A third route has matured in recent years. The collision-free configuration space $\mathcal{C}_{\text{free}}$ can be approximated as a union of convex polytopes [10]–[14], with seeding automated through clique covers of visibility graphs [15]. On such a decomposition, Graphs of Convex Sets (GCS) [16], [17] compute trajectories that are globally optimal *relative to the decomposition* by solving a tight convex relaxation of a mixed-integer program. These tools turn $\mathcal{C}_{\text{free}}$ into an object that an optimiser can reason about directly, but they have so far mainly been applied to collision-free motion of the robot alone. [CHECK: Check whether any recent work applies C-free decompositions + GCS to prehensile regrasp planning; closest known: contact-rich planar pushing [18], GCS* [19].]

In this paper, we bring convex decompositions to prehensile manipulation planning. We decompose the collision-free composite configuration space, describe the grasp and placement spaces as unions of polytopes, and intersect them with the decomposition. The nodes of the classical manipulation

This work was supported by the UKRI AI Centre for Doctoral Training in Dependable and Deployable AI for Robotics (CDT-D2AIR) [TODO: grant number].

graph, i.e., the connected components of grasp-and-placement configurations, then become unions of convex sets over which we can optimise. A minimum-regrasp sequence of grasp and release configurations is selected by a mixed-integer quadratic program (MIQP) or by A* search over the polytope graph [20], and connected by optimal transit and transfer trajectories computed with GCS (Fig. 1). A preliminary version of this pipeline was demonstrated on 3-DOF tasks in [21]. **[CHECK: anonymise if the venue is double-blind.]**

Scaling this idea to a full arm raises a problem that does not appear in low-dimensional examples. For a gripper with a few joints, grasp and placement spaces can be written directly as bounds on the configuration. For a 7-DOF arm, a grasp is a nonlinear constraint on the forward kinematics, and the set of grasping configurations is a thin, curved, lower-dimensional manifold. To represent it with polytopes, we use NLP sampling [22], a restarting two-phase constrained sampler, to generate diverse grasp configurations, and construct around each one a polytope aligned with the null space of the active grasp constraints.

Our contributions are:

- A formulation of prehensile manipulation planning over convex decompositions of the collision-free composite configuration space, in which the grasp, placement and transition sets of the manipulation graph are represented as unions of convex polytopes (Sections III and IV-D).
- A two-stage planner that selects a minimum-regrasp sequence of grasp and release configurations, by MIQP or by A* search over the polytope graph, and connects it with GCS transit and transfer trajectories that are optimal relative to the decomposition (Sections IV-E–IV-F).
- A construction of grasp polytopes for high-DOF arms, combining NLP sampling of the nonlinear grasp constraints with null-space-aligned polytopes around the samples (Section IV-C).
- An evaluation on bottle-cap opening tasks requiring repeated regrasping, from 3-DOF and 4-DOF systems to the 10-dimensional composite space of a Franka Panda arm, hand and cap, with a comparison against sampling-based planners (Sections V–VI). **[TODO: update once the baselines are fixed.]**

II. RELATED WORK

A. Manipulation and multi-modal planning

Manipulation planning extends motion planning with movable objects that can only move when grasped [1], [23]. Early methods assumed finite, user-given sets of grasps and placements, and searched over the resulting discrete transitions [1], [24]. Treating grasps and placements as continuous sets was a major step: in the manipulation graph of Siméon et al. [2], nodes are connected components of the intersection of grasp and placement spaces and edges are transit or transfer paths, both computed with probabilistic roadmaps. Multi-modal planners generalise this structure to sequences of modes connected through their intersections [5],

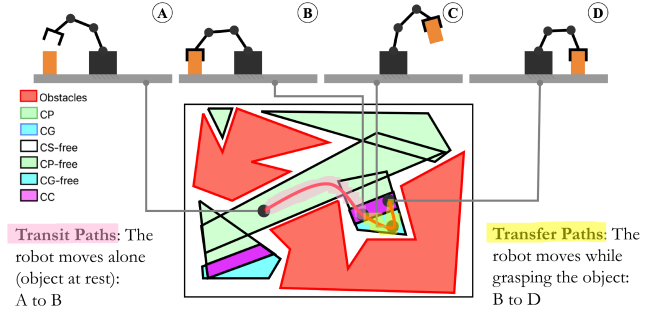


Fig. 1. Manipulation planning in the composite configuration space of robot and object. Valid states lie in the placement space $\mathcal{C}_P$ (object at rest, green) or the grasp space $\mathcal{C}_G$ (object held, blue), intersected with a convex decomposition of the collision-free space. Their intersection (purple) contains the configurations where the robot can grasp or release the object. A plan alternates transit paths in $\mathcal{C}_P$ (A→B) and transfer paths in $\mathcal{C}_G$ (B→D). **[TODO: replace with a figure showing the real Franka/cap task alongside the abstract picture, and re-label CC as $\mathcal{C}_G \cap \mathcal{C}_P$.]**

[25], and asymptotically optimal variants exist [7], [26]. Task and motion planning (TAMP) adds symbolic decisions on top of this geometry [27], either by calling continuous samplers from a symbolic planner [28] or by optimising over a fixed skeleton [29]. Constraint-based frameworks such as HPP represent grasp, placement and motion constraints explicitly and explore their connectivity with constrained solvers [30]. Our planner keeps the manipulation-graph structure of [2], but represents its nodes and edges with convex sets rather than samples, so that the choice of grasps and the motions between them can be optimised.

B. Convex decompositions of C-free and Graphs of Convex Sets

Describing configuration-space obstacles exactly is intractable [31], [32], which motivates describing $\mathcal{C}_{\text{free}}$ directly as a union of simpler sets. IRIS grows large convex regions in free space by alternating separating hyperplanes and inscribed ellipsoids [10]. C-IRIS certifies regions as collision-free through a rational parametrisation of the kinematics and sums-of-squares programs [11], [12], while IRIS-NP, IRIS-NP2 and IRIS-ZO trade certification for speed and scale with probabilistic guarantees [13], [14]. Clique covers of visibility graphs automate seeding and cover $\mathcal{C}_{\text{free}}$ with few regions [15], and recent GPU implementations make decompositions fast enough for online use [33]. Convex decompositions have long been used in mixed-integer planning [34], [35], and also in contact planning for legged robots [36]–[38]. GCS [16], [17] solves shortest-path problems over such decompositions through a tight convex relaxation, and has been extended to non-Euclidean joint spaces [39], temporal-logic specifications [40], contact-rich planar manipulation [18], and heuristic search over implicitly defined graphs (GCS*) [19]. Interleaving graph search with convex or trajectory optimisation has also been explored in [41]. To our knowledge, these tools have not been used to plan grasp and regrasp sequences for

prehensile manipulation in the composite configuration space. **[CHECK: novelty claim]**

C. Constrained sampling

Grasps and placements define lower-dimensional constraint manifolds. Task space regions express pose constraints for sampling-based planners [42], and manifold planners explore such constraints through tangent-space steps followed by projection [43], [44]; see [45] for a review. Since these methods expand continuously from known feasible states, they struggle when the feasible set is disconnected. NLP sampling [22] instead targets diverse samples from densities restricted to non-linear equality and inequality constraints, through a restarting two-phase scheme: a Gauss–Newton phase that finds a feasible point, and an interior phase (e.g., nonlinear hit-and-run or manifold RRT) that explores the feasible set from it. We use this sampler to seed grasp polytopes on the grasp manifold of a 7-DOF arm.

III. PROBLEM FORMULATION

A. Composite configuration space

We consider a robot with configuration space $\mathcal{C}_{\text{rob}} \subset \mathbb{R}^{n_r}$ and a movable rigid object with configuration space $\mathcal{C}_{\text{obj}} \subset \mathbb{R}^{n_o}$, in a known static environment. The state of the whole system lives in the composite configuration space

$$\mathcal{C} = \mathcal{C}_{\text{rob}} \times \mathcal{C}_{\text{obj}} \subset \mathbb{R}^n, \quad n = n_r + n_o, \quad (1)$$

and we write $\mathcal{C}_{\text{free}} \subseteq \mathcal{C}$ for the configurations in which no pair of bodies (robot links, object, obstacles) collides, apart from the contacts required by a grasp. We restrict the robot and object to revolute and prismatic joints, with finite joint limits $q_{\min} \leq q \leq q_{\max}$; we write q_{cap} for the object coordinate in the tasks below. **[CHECK: the MInf report argues other joint types reduce to revolute+prismatic [46]; keep or drop.]**

B. Placement and grasp spaces

The object cannot move by itself: at every instant it is either resting at a stable placement or held by the robot. Following [2], we define two subsets of $\mathcal{C}$:

- the *placement space* $\mathcal{C}_P$, the configurations in which the object rests at a stable placement;
- the *grasp space* $\mathcal{C}_G$, the configurations in which the robot holds the object with a valid grasp.

Every valid configuration lies in $\mathcal{C}_P \cup \mathcal{C}_G$. Configurations in $\mathcal{C}_G \cap \mathcal{C}_P$, where the object is both grasped and stably placed, are those at which the robot can grasp or release the object.

C. Transit, transfer and manipulation constraints

A manipulation path alternates two kinds of motion:

- *Transit*: the robot moves alone in $\mathcal{C}_P$ while the object stays at a fixed placement, i.e., the object coordinates are constant.
- *Transfer*: the robot moves in $\mathcal{C}_G$ while holding the object, so that the relative pose between end-effector and object is constant.

We collect these requirements in a set of *manipulation constraints* $\mathcal{M}$. For two configurations q, q' at the endpoints of a segment, we write $\mathcal{M}_{\text{tr}}(q, q')$ for the transit constraint and $\mathcal{M}_{\text{tf}}(q, q')$ for the transfer constraint. In general, $\mathcal{M}_{\text{tf}}$ is a nonlinear function of the forward kinematics. In the bottle-cap tasks of Section V, the cap only rotates about its own axis, which is aligned with the axis of the last robot joint during a grasp. Both constraints are then linear:

$$\mathcal{M}_{\text{tr}}(q, q') : \quad q_{\text{cap}} = q'_{\text{cap}}, \quad (2)$$

$$\mathcal{M}_{\text{tf}}(q, q') : \quad q_{\text{cap}} - q_{\text{wr}} = q'_{\text{cap}} - q'_{\text{wr}}, \quad (3)$$

where q_{wr} is the coordinate of the joint that rotates the gripper about the cap axis, together with the requirement that q and q' lie in the same connected grasp set (Section IV-D). **[CHECK: axis-sign convention: the MInf report notes the cap and gripper axes are inverted and aligned w.l.o.g.]**

D. Problem statement

Given $\mathcal{C}$, $\mathcal{C}_P$, $\mathcal{C}_G$, the manipulation constraints $\mathcal{M}$, and initial and goal configurations $q_{\text{init}}, q_{\text{goal}} \in \mathcal{C}_P$, find a continuous trajectory $\tau : [0, 1] \rightarrow \mathcal{C}_{\text{free}}$ with $\tau(0) = q_{\text{init}}$ and $\tau(1) = q_{\text{goal}}$ that is a concatenation of segments

$$\tau = \tau_0 \oplus \tau_1 \oplus \dots \oplus \tau_{2\ell}, \quad (4)$$

where even segments τ_{2i} are transit paths in $\mathcal{C}_P \cap \mathcal{C}_{\text{free}}$ and odd segments τ_{2i+1} are transfer paths in $\mathcal{C}_G \cap \mathcal{C}_{\text{free}}$, each satisfying its manipulation constraint along its whole length. Among such trajectories, we seek one with the minimum number of grasps ℓ and, for that ℓ , minimum length.

The breakpoints of τ form a sequence of *keyframes*

$$x_0 = q_{\text{init}}, \quad x_1 = g_1, \quad x_2 = r_1, \quad \dots, \quad x_{2\ell} = r_\ell, \quad x_{2\ell+1} = q_{\text{goal}}, \quad (5)$$

where g_i and r_i are the i -th grasp and release configurations. Every grasp and release lies in $\mathcal{C}_G \cap \mathcal{C}_P$; the pairs (x_{2i}, x_{2i+1}) are connected by transit paths and the pairs (x_{2i+1}, x_{2i+2}) by transfer paths. Our planner first chooses the keyframes and then the paths between them.

IV. METHOD

A. Overview

Algorithm 1 summarises the planner. An offline stage decomposes $\mathcal{C}_{\text{free}}$ into convex polytopes (Section IV-B), constructs polytopic grasp and placement spaces (Section IV-C), and intersects them to obtain the convex counterparts of the nodes and edges of the manipulation graph (Section IV-D). The planning stage selects the keyframes (5) with the minimum number of grasps (Section IV-E) and connects them with transit and transfer trajectories (Section IV-F). The offline stage depends only on the robot, object and environment, and can be reused across queries with different q_{init} and q_{goal} .

Algorithm 1 Manipulation planner

Input: robot, object, obstacles, manipulation constraints $\mathcal{M}$, q_{init} , q_{goal} , max. grasps L

Output: trajectory τ

```

1:  $\{P_j\}_{j=1}^J \leftarrow \text{CLIQUECOVERIRIS}(\mathcal{C}_{\text{free}})$   $\triangleright$  Sec. IV-B
2:  $\{G_k\}_{k=1}^K \leftarrow \text{GRASPPOLYTOPES}()$   $\triangleright$  Sec. IV-C, Alg. 2
3:  $\mathcal{C}_P^{\text{free}}, \mathcal{C}_G^{\text{free}}, \text{components} \leftarrow \text{INTERSECT}(\{P_j\}, \{G_k\}, \mathcal{C}_P)$ 
    $\triangleright$  Sec. IV-D
4: for  $\ell = 1, \dots, L$  do
5:    $x_{0:2\ell+1} \leftarrow \text{SEQUENCE}(\ell, \mathcal{C}_P^{\text{free}}, \mathcal{C}_G^{\text{free}}, \mathcal{M})$   $\triangleright$  Sec. IV-E
6:   if a sequence is found then break
7: for  $i = 0, \dots, 2\ell$  do
8:    $\sigma \leftarrow \text{transit if } i \text{ even, else transfer}$ 
9:    $\tau_i \leftarrow \text{GCS}(x_i, x_{i+1}; \text{regions of mode } \sigma, \mathcal{M}_\sigma)$   $\triangleright$ 
   Sec. IV-F
10: return  $\tau = \tau_0 \oplus \dots \oplus \tau_{2\ell}$ 

```

B. Convex decomposition of the collision-free space

We compute a collection of convex polytopes $P_j = \{q \in \mathbb{R}^n : A_j q \leq b_j\}$, $j = 1, \dots, J$, whose union $\mathcal{P} = \bigcup_j P_j$ is an inner approximation of $\mathcal{C}_{\text{free}}$. We use clique covers of visibility graphs [15]: collision-free configurations are sampled, a visibility graph is built between them, it is covered by cliques, and each clique seeds the growth of one region, here with IRIS-NP [13] as implemented in Drake [47]. The process repeats until a fraction α of sampled collision-free configurations is covered. Regions grown this way are collision-free with high probability; when certification is required, they can be mapped to the tangent configuration space $t_i = \tan(\theta_i/2)$ of each revolute joint and certified, or shrunk until certified, with C-IRIS [12]. **[CHECK: the MInf pipeline maps polytope vertices to TC-space, takes the convex hull, shrinks with C-IRIS and maps vertices back; the back-mapped hull is not guaranteed to stay inside the certified set, since the map is nonlinear. Either fix, state as approximate, or drop certification from the 10-DOF results.]** The decomposition is computed once per scene, and is the only stage whose cost grows directly with the volume of free space covered. **[TODO: coverage α and IRIS/clique-cover parameters per task; the stored 10-DOF decomposition has 18 regions at $\alpha = 0.98$ [CHECK:].]**

C. Grasp and placement polytopes

The planner needs $\mathcal{C}_P$ and $\mathcal{C}_G$ as unions of polytopes, $\mathcal{C}_P = \bigcup_{k'} P_{k'}^{\text{pl}}$ and $\mathcal{C}_G \supseteq \bigcup_k G_k$. In the bottle-cap tasks, the cap rotates about a fixed axis and is always stably placed, so $\mathcal{C}_P$ is the joint-limit box. Grasps are where the two settings differ.

a) *Low-DOF grippers:* When the gripper joints map directly to the grasp, $\mathcal{C}_G$ is a box in configuration space. For the 3- and 4-DOF tasks of Section V, a grasp is any configuration in which the finger touches the cap, $z_{\text{finger}} = -0.025$ m, with the other joints free within their limits. To give the set non-zero volume we use a 1 mm band, $z_{\text{finger}} \in [-0.025, -0.024]$, so $\mathcal{C}_G$ is a single box, i.e., the convex hull of its 2^n corners.

Algorithm 2 Grasp polytopes for a high-DOF arm

Input: grasp constraints h, g and Jacobians; joint limits; margins ϵ , tolerance η

Output: grasp polytopes $\{G_k\}$

```

1:  $\{q_k\} \leftarrow \text{NLPSAMPLING}(h, g)$   $\triangleright$  diverse grasp seeds
2: for each seed  $q_k$  do
3:    $J_k \leftarrow [\partial h / \partial q; \partial g_A / \partial q](q_k)$   $\triangleright$  active rows
4:    $N_k \leftarrow \text{null}(J_k)$ ,  $V_k \leftarrow N_k^\perp$   $\triangleright$  orthonormal bases
5:    $r_k \leftarrow \text{largest extents along } N_k \text{ s.t. the box validates}$ 
6:    $G_k \leftarrow \{q : |N_k^\top (q - q_k)| \leq r_k, |V_k^\top (q - q_k)| \leq \epsilon\}$ 
7:   validate  $G_k$ :  $\|h\|_\infty \leq \eta$ ,  $g \leq \eta$  on corners and random
   points

```

b) *High-DOF arms:* For a 7-DOF arm, a grasp constrains the pose of the hand frame E relative to the cap frame C , ${}^C X^E(q) = ({}^C R^E(q), {}^C p^E(q))$, which is a nonlinear function of the arm joints. For the cap task, the hand must be centred above the cap, within a height band, with its approach axis $z_E = {}^C R^E(q) e_z$ pointing down along the cap axis:

$$h(q) = [p_x, p_y, z_{E,x}, z_{E,y}]^\top = 0, \quad (6)$$

$$g(q) = [p_z - z_{\text{max}}, z_{\text{min}} - p_z, z_{E,z} + c]^\top \leq 0, \quad (7)$$

with $p = {}^C p^E(q)$, $[z_{\text{min}}, z_{\text{max}}] = [0.105, 0.110]$ m and $c = 0.9$, together with the finger joints restricted to a 1 mm band around the cap width. The set $\{q : h(q) = 0, g(q) \leq 0\}$ is a thin, curved manifold of dimension $n - 4$ in $\mathcal{C}$, which cannot be written down as a polytope. We approximate it with a union of thin polytopes around diverse grasp configurations (Algorithm 2).

c) *Grasp seeds with NLP sampling:* Grasp polytopes are only useful if their seeds cover the different components of the grasp manifold (e.g., different elbow postures). A local inverse-kinematics solver returns one solution per initial guess and gives no control over diversity. We instead use NLP sampling [22], which targets the density $p(q) \propto e^{-f(q)} \mathbb{1}_{[g(q) \leq 0]} \mathbb{1}_{[h(q) = 0]}$, here with $f \equiv 0$. Each restart (i) picks a seed among candidates far from all previous samples, (ii) runs a damped Gauss–Newton descent on the slack $\|s(q)\|^2$, $s = [\max(g, 0); h]$, until a feasible point is found, and (iii) explores the feasible set from it with an interior sampler, relaxing $h = 0$ to a tube $|h| \leq \eta_w$ and projecting proposal directions onto the tangent space, $d \leftarrow (I - J_h^\top (J_h J_h^\top + \lambda I)^{-1} J_h) d$. A final Gauss–Newton projection brings every sample within $\|h\|_\infty \leq 10^{-3}$. Our implementation is a Python port of the reference implementation of [22], with nonlinear hit-and-run (NHR) and manifold RRT (mRRT) as interior methods, and Drake’s automatic differentiation for the Jacobians. On the grasp problem, NHR mixed poorly in the wrist and cap coordinates: its line search shares one step size across all coordinates, so the tightest constraint throttles motion even along directions it does not depend on. We diagnose this by comparing the spread of each coordinate within a chain with its spread across restarts, and switch to mRRT, which takes fixed-length tangent steps and corrects them afterwards. **[TODO:**

report within-chain vs. across-restart spread for HR vs. mRRT; number of restarts / samples / seeds used; optionally MSTs curves [22].]

d) *Null-space polytopes*: Around each seed q_k , we linearise the active constraints, i.e., h and the rows of g within δ of zero, and compute an orthonormal basis N_k of the null space of their Jacobian J_k and a basis V_k of its orthogonal complement. Directions in N_k preserve the grasp to first order, and some of them preserve it exactly: rotating the wrist, whose axis passes through the origin of E along z_E , changes neither p nor z_E ; the cap joint and the fingers do not enter (6)–(7) at all. These directions are given their full joint range. The remaining null direction of the 7-DOF arm, a self-motion that mixes all shoulder and elbow joints, is given the largest extent for which the constraints remain within a tolerance η , and the complement V_k a small margin ϵ . The resulting G_k is a box in rotated coordinates, i.e., an H-polytope with $2n$ half-spaces, intersected with the joint limits. We validate each G_k by checking the constraints at the corners of the non-trivial directions and at random interior points. **[TODO: final ϵ and η (currently $\epsilon = 10^{-3}$ rad, matching the finger band, and $\eta = 10^{-2}$); report validation counts and volume compared with the axis-aligned box ($\pm 3.5 \times 10^{-3}$ rad, preliminary) [CHECK: numbers from the investigation record on branch nlp-sampling/quality-metrics].] [CHECK: this validation is empirical, not a certificate; say so explicitly or add a Lipschitz/interval bound.]**

Because the wrist and cap are exact invariant directions of the grasp constraint, a transfer motion for the cap task (a coupled rotation of wrist and cap, (3)) stays within a single G_k ; regrasps between different grasp polytopes happen through transit motions.

D. Mode sets and their connectivity

We intersect the grasp and placement spaces with the decomposition,

$$\mathcal{C}_P^{\text{free}} = \bigcup_j (\mathcal{C}_P \cap P_j), \quad \mathcal{C}_G^{\text{free}} = \bigcup_{j,k} (G_k \cap P_j), \quad (8)$$

keeping only non-empty intersections (each checked with one linear program). Both are unions of convex polytopes, possibly disconnected. We compute their connected components from the intersection graph of their polytopes: two polytopes are adjacent if they intersect. The components of $\mathcal{C}_P^{\text{free}}$ and $\mathcal{C}_G^{\text{free}}$ play the role of the transit and transfer edges of the manipulation graph: two configurations in the same component of $\mathcal{C}_P^{\text{free}}$ can be connected by a collision-free path in $\mathcal{C}_P$, and similarly for $\mathcal{C}_G^{\text{free}}$. Configurations in $\mathcal{C}_G^{\text{free}} \cap \mathcal{C}_P^{\text{free}}$ are candidate grasps and releases, the nodes of the manipulation graph [2].

E. Grasp–release sequencing

a) *Mixed-integer formulation*: For a fixed number of grasps ℓ , we choose the keyframes (5) by solving

$$\begin{aligned} \min_{x_0, \dots, x_{2\ell+1}} \quad & \sum_{i=0}^{2\ell} \|x_{i+1} - x_i\|_2^2 \\ \text{s.t.} \quad & x_0 = q_{\text{init}}, \quad x_{2\ell+1} = q_{\text{goal}}, \quad q_{\min} \leq x_i \leq q_{\max}, \\ & \mathcal{M}_{\text{tr}}(x_{2i}, x_{2i+1}), \quad x_{2i} \sim_{\mathcal{C}_P^{\text{free}}} x_{2i+1}, \quad i \leq \ell, \\ & \mathcal{M}_{\text{tf}}(x_{2i+1}, x_{2i+2}), \quad x_{2i+1} \sim_{\mathcal{C}_G^{\text{free}}} x_{2i+2}, \quad i < \ell, \end{aligned} \quad (9)$$

where $x \sim_{\mathcal{U}} x'$ requires both points to lie in the same connected component of the union of polytopes $\mathcal{U}$. For $\mathcal{U} = \bigcup_m Q_m$, $Q_m = \{x : A_m x \leq b_m\}$, with components $\mathcal{U}_1, \dots, \mathcal{U}_c$, we encode membership of each point x with binaries δ_m (point in Q_m) and λ_c (point in $\mathcal{U}_c$):

$$\begin{aligned} A_m x &\leq b_m + M(1 - \delta_m), \quad \sum_m \delta_m \geq 1, \\ \sum_c \lambda_c &= 1, \quad \delta_m \leq \lambda_{c(m)}, \end{aligned} \quad (10)$$

and require the component indicators λ of the two points of a pair to be equal. Since every grasp and release keyframe belongs to a transit pair and a transfer pair, it lies in $\mathcal{C}_G^{\text{free}} \cap \mathcal{C}_P^{\text{free}}$. The objective, a sum of squared keyframe distances, is a convex surrogate for the length of the final trajectory. Problem (9) is a mixed-integer convex quadratic program, which we solve with MOSEK [48] for $\ell = 1, 2, \dots$ until the first feasible ℓ , which gives the minimum number of grasps (relative to the decomposition, see Proposition 1).

On the 10-DOF task, the MIQP fails to return feasible sequences even where they exist. We attribute this to numerics: the grasp polytopes have widths of order $\epsilon = 10^{-3}$, while the big-M constant must dominate the constraint violation over the whole joint range, and the problem grows with $(2\ell + 2)$ copies of all region binaries. **[CHECK: confirm cause; possible fixes: per-row tight M from joint limits, convex-hull (perspective) formulation, rescaling the thin directions.]** This motivates the graph-search alternative below.

b) *A* search over the polytope graph*: **[TODO: Draft — align with the implementation in progress.]** We build a graph whose vertices are the polytopes of $\mathcal{C}_P^{\text{free}}$ (transit vertices) and of $\mathcal{C}_G^{\text{free}}$ (transfer vertices). Two vertices of the same mode are adjacent if their polytopes intersect; a transit and a transfer vertex are adjacent if their intersection contains a configuration in $\mathcal{C}_G \cap \mathcal{C}_P$ (a grasp or release). A search node stores a vertex, the mode, an entry configuration q and the number of grasps so far. Expanding a node towards a neighbour solves a small convex QP for the entry configuration q' closest to q in the intersection of the two polytopes subject to the manipulation constraint of the current mode, so that keyframes remain consistent with (2)–(3). A* orders nodes by $f = g + h$, with g the accumulated keyframe distance plus a large cost per grasp, which makes the number of grasps the primary criterion, and h an admissible lower bound, e.g., $\|q' - q_{\text{goal}}\|$ plus the grasp cost times $\lceil |q'_{\text{cap}} - q_{\text{cap,goal}}| / (q_{\text{wr,max}} - q_{\text{wr,min}}) \rceil$, a lower bound on the remaining number of grasps for the cap task. Fixing entry configurations greedily at expansion keeps each step cheap but gives up optimality and completeness relative to the

decomposition; GCS* [19] recovers both by keeping a convex program over the whole partial path and pruning dominated paths, at a higher cost per expansion. **[TODO: choose between greedy entry points and a GCS*-style expansion; state the heuristic used and whether it is admissible.]**

F. Transit and transfer trajectories with GCS

Given the keyframes, each pair (x_i, x_{i+1}) is connected by a shortest path in the graph of convex sets formed by the polytopes of the corresponding mode, $\mathcal{C}_P^{\text{free}}$ for transit and $\mathcal{C}_G^{\text{free}}$ for transfer, using Drake's `GcsTrajectoryOptimization` [16], [47]. Each trajectory segment is a Bézier curve whose control points lie in one polytope, so the whole curve lies in that polytope and in $\mathcal{P}$. When the manipulation constraints are linear, as in (2)–(3), we impose them on all control points, and since a Bézier curve is a convex combination of its control points, they then hold along the entire segment. **[CHECK: the MInf version did not enforce $\mathcal{M}$ inside GCS (only at keyframes); confirm this is now implemented.]** The final trajectory is the concatenation $\tau = \tau_0 \oplus \dots \oplus \tau_{2\ell}$. **[TODO: Bézier order, cost weights (path length, time), convex relaxation and rounding settings.]**

G. Guarantees

Proposition 1 (informal): Let $\mathcal{P} \subseteq \mathcal{C}_{\text{free}}$. (i) Any trajectory returned by Algorithm 1 is collision-free, respects joint limits and satisfies the linear manipulation constraints along every segment. (ii) Problem (9) is a relaxation of the planning problem restricted to the represented sets: every valid manipulation path with ℓ grasps whose transit segments lie in $\mathcal{C}_P^{\text{free}}$ and transfer segments in $\mathcal{C}_G^{\text{free}}$ yields a feasible keyframe sequence for (9). Hence, if (9) is infeasible for all $\ell \leq L$, no such path with at most L grasps exists within the decomposition, and if the GCS stage succeeds for the first feasible ℓ , the plan uses the minimum number of grasps relative to the decomposition. *Sketch.* (i) follows from the containment of each Bézier segment in a polytope of $\mathcal{P}$ and the convex-combination argument above. (ii) The keyframes of a valid path satisfy the constraints of (9), since consecutive keyframes of a transit (transfer) segment are connected within $\mathcal{C}_P^{\text{free}}$ ($\mathcal{C}_G^{\text{free}}$) and therefore lie in the same component. The converse does not hold: two configurations in the same component of $\mathcal{C}_P^{\text{free}}$ need not be connected by a path that keeps q_{cap} fixed, so the GCS stage can fail on a feasible sequence. **[CHECK: state precisely; relates to the 4-DOF, $\alpha = 0.90$ case in Table I.]** Optimality of the path length is only partial: each segment is optimal relative to the decomposition given its endpoints when the convex relaxation of GCS is tight [16], [17], but the keyframes are chosen with a surrogate cost. $\square$

V. EXPERIMENTAL SETUP

A. Tasks

All tasks are variants of a simplified bottle-cap opening task: a cap rotates about its fixed vertical axis, and the goal is to turn it by a full revolution, from $q_{\text{cap}} = -3.14$ to $q_{\text{cap}} = 3.14$ rad. The joint that rotates the gripper about the cap axis has a

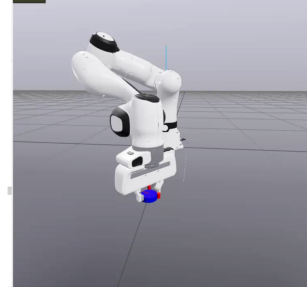


Fig. 2. Task T4: a Franka Panda grasping the bottle cap, a configuration sampled on the grasp manifold (6)–(7). **[TODO: replace with snapshots of a full plan with obstacles visible, and a low-dimensional slice of $\mathcal{C}_G^{\text{free}}$ / $\mathcal{C}_P^{\text{free}}$.]**

limited range, so the task cannot be completed with a single grasp. Since q_{cap} is constant during transit (2) and changes as much as q_{wr} during transfer (3), a range of 2 rad requires at least four grasps. **[CHECK: verify against the plans found.]** The task is deliberately simple in its object motion, so that difficulty comes from regrasping, obstacles near the contact, and the dimension of the composite space.

- **T1, 3-DOF.** A simplified Schunk WSG gripper welded at a fixed distance from the cap, with a revolute joint about the cap axis, $\theta_{\text{gr}} \in [-1, 1]$ rad, and one prismatic finger, $z_{\text{finger}} \in [-0.055, 0]$ m; the composite configuration is $q = (q_{\text{cap}}, \theta_{\text{gr}}, z_{\text{finger}})$. Two cylindrical obstacles are placed next to the cap. Its 3D composite space can be visualised directly.
- **T2, 4-DOF.** T1 with the second finger as an additional prismatic joint.
- **T3, Franka hand, 4-DOF.** **[CHECK: keep?]** A Franka Panda with all arm joints but the wrist locked, two fingers and the cap.
- **T4, Franka arm, 10-DOF.** A Franka Panda [49] with its 7 arm joints and 2 finger joints, and the cap, placed 0.5 m in front of the robot base, with two cylindrical obstacles next to it (Fig. 2). The wrist joint is restricted to $[-1, 1]$ rad. Grasps are defined by (6)–(7). **[TODO: obstacle sizes; start/goal arm configurations.]**

B. Implementation

The planner is implemented in Python on top of Drake [47] (v1.48.0; later versions showed a regression with the MOSEK interface), with MOSEK [48] solving the MIQP and the GCS programs. The clique-cover decompositions use Drake's IRIS-NP implementation. The NLP sampler is our Python port of the reference implementation of [22], with forward kinematics and Jacobians from Drake's automatic differentiation. **[TODO: hardware; code release link.]**

C. Baselines and ablations

[TODO: Decide and implement. Candidates:]

- a sampling-based manipulation planner over the manipulation graph [2], e.g., in HPP [30];

TABLE I
COVERAGE OF THE DECOMPOSITION ON THE LOW-DOF TASKS [TODO: PRELIMINARY, EARLIER PIPELINE]

Task	Coverage target (actual)	Regions	Decomp. (s)	Traj. (s)
T1	0.70–0.90 (0.94)	4	29.8	29.5
	0.96 (0.98)	7	54.6	34.0
	0.99 (0.996)	11	87.7	112.6
T2	0.70 (0.77)	4	16.5	— ^a
	0.90 (0.95)	5	31.2	— ^b
	0.96 (0.98)	10	78.5	29.0
	0.99 (0.996)	14	115.9	95.9

^aNo feasible grasp–release sequence. ^bSequence found, transit/transfer connection failed.

- constrained sampling-based planners from OMPL [44], [50] for the transit and transfer segments, given the same keyframes;
- ablations: MIQP vs. A* sequencing; axis-aligned vs. null-space grasp polytopes; NHR vs. mRRT seeds; coverage α .

D. Metrics

We report success rate over [TBD] queries and random seeds; offline time (decomposition and grasp polytopes); on-line time (sequencing and GCS); number of grasps; trajectory length in configuration space; and the maximum violation of the grasp constraint $\|h\|_\infty$ along transfer segments, a measure of how faithfully the polytopes represent the grasp manifold.

VI. RESULTS

A. Low-DOF tasks

Table I reports the effect of the coverage of the decomposition on the low-DOF tasks. [TODO: These numbers are from the MInf report / Year-1 annual review (earlier pipeline, no mode constraints inside GCS). Re-run with the final pipeline, add T3, and average over seeds.] In T1, even moderate coverage is enough to find a plan, but trajectory time grows with the number of regions. In T2, low coverage is not enough: at a target of 0.70 no feasible grasp–release sequence exists in the decomposition, and at 0.90 a sequence is found but its transit and transfer segments cannot be connected. Full plans are found from a target coverage of about 0.96. This is expected, as grasps lie close to contact, near the boundary of $\mathcal{C}_{\text{free}}$, and small gaps in the decomposition can remove exactly the regions needed to regrasp. Trajectory time depends on the number and shape of the regions as much as on the dimension: T1 at 0.99 takes longer than T2 at 0.96.

B. 10-DOF Franka task

[TODO: Main results table: rows = ours (MIQP), ours (A*), baselines; columns = success, offline time, online time, grasps, length, max $\|h\|_\infty$ along transfers.]

TABLE II
RESULTS ON T4 (10-DOF) [TODO: PLACEHOLDER]

Method	Succ.	Offline (s)	Online (s)	Grasps	Length
Ours (A*)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Ours (MIQP)	[TBD]	[TBD]	[TBD]	[TBD]	[TBD]
Sampling-based baseline	[TBD]	—	[TBD]	[TBD]	[TBD]

C. Grasp polytopes and seeds

[TODO: Number of NLP samples, restarts and seeds; HR vs. mRRT mixing diagnostic; number and volume of grasp polytopes; validation failures; max $\|h\|_\infty$ inside the polytopes; axis-aligned vs. null-space ablation.]

VII. DISCUSSION

a) *Coverage near contact:* The low-DOF experiments show that the planner is only as reliable as the decomposition it operates on. Uniform coverage of $\mathcal{C}_{\text{free}}$ is not what matters most: grasps happen near contact, at the boundary of $\mathcal{C}_{\text{free}}$, and missing a small region there can make an otherwise solvable task infeasible. This suggests biasing the decomposition towards task-relevant sets, e.g., by seeding region growth with samples on the grasp and placement manifolds rather than uniformly. [TODO: revisit once 10-DOF coverage results are in.]

b) *Representing grasp manifolds with polytopes:* Before the null-space construction of Section IV-C, we explored two alternatives on T4. First, we used NLP samples of the grasp constraints as seeds for IRIS-type region growing. This produced a very large number of very small polytopes, since regions grown around a thin curved manifold are limited by its curvature in every direction, which made sequencing slow and fragmented. Second, we fitted a union of polytopes to the NLP samples with a small max–min network, in which each neuron is a half-space $a_k^\top q - b_k$, a region is the maximum over its half-spaces and the set is the minimum over regions, with unused half-spaces pruned. This learns all polytopes at once, but did not yet represent the grasp set accurately enough to plan with. [TODO: one figure or numbers for each, if space allows.] The null-space construction works because it uses the structure of the constraint, in particular its exact invariance to wrist and cap rotation, instead of discovering it from samples. Its cost is that each polytope is thin, which affects the numerics of the MIQP (Section IV-E).

c) *Limitations:* The method addresses prehensile manipulation of a single rigid object with known geometry, quasi-statically. Linear manipulation constraints (2)–(3) follow from the symmetry of the cap task; general objects with $SE(3)$ placements make the transfer constraint nonlinear, and would require either polytopes in grasp-relative coordinates or constraints enforced approximately. The grasp polytopes are validated empirically rather than certified, and the decomposition is collision-free only with high probability unless certified. GCS returns solutions that are optimal relative to the decomposition only when its convex relaxation is tight [17],

and the keyframes are chosen with a surrogate cost. Finally, the offline decomposition is expensive in high dimension and must be recomputed when the scene changes, although GPU implementations [33] reduce this cost.

d) *Future work*: Natural extensions are constrained sampling inside region growing, so that regions are grown directly on the grasp manifold (we have ported IRIS-ZO and clique covers to Python to make the particle sampler replaceable), richer contact modes such as pushing, pivoting and in-hand manipulation, and experiments on the real robot. **[TODO: trim to what fits.]**

VIII. CONCLUSION

We presented a manipulation planner that operates on convex decompositions of the collision-free composite configuration space of robot and object. By representing the grasp and placement spaces as unions of polytopes and intersecting them with the decomposition, the nodes and edges of the classical manipulation graph become convex sets. Grasp and release configurations are then selected with the minimum number of grasps by mixed-integer optimisation or heuristic search, and connected by transit and transfer trajectories computed with Graphs of Convex Sets. To scale this representation to a 7-DOF arm, where grasping is a nonlinear constraint on the forward kinematics, we combined NLP sampling for diverse grasp configurations with polytopes aligned with the null space of the grasp constraints. **[TODO: one or two sentences on the 10-DOF results and the comparison with sampling-based planners.]** More broadly, this work shows that describing where grasping actions can start and end as convex sets lets an optimiser reason about how to sequence them.

ACKNOWLEDGMENT

[TODO: acknowledgments; funding is in the title footnote.]

REFERENCES

- [1] R. Alami, T. Siméon, and J.-P. Laumond, “A geometrical approach to planning manipulation tasks. the case of discrete placements and grasps,” in *The Fifth International Symposium on Robotics Research*. MIT Press, 1990, pp. 453–463.
- [2] T. Siméon, J.-P. Laumond, J. Cortés, and A. Sahbani, “Manipulation planning with probabilistic roadmaps,” *The International Journal of Robotics Research*, vol. 23, no. 7–8, pp. 729–746, 2004.
- [3] L. E. Kavraki, P. Švestka, J.-C. Latombe, and M. H. Overmars, “Probabilistic roadmaps for path planning in high-dimensional configuration spaces,” *IEEE Transactions on Robotics and Automation*, vol. 12, no. 4, pp. 566–580, 1996.
- [4] S. M. LaValle, “Rapidly-exploring random trees: A new tool for path planning,” Iowa State University, Tech. Rep. TR 98-11, 1998.
- [5] K. Hauser and J.-C. Latombe, “Multi-modal motion planning in non-expansive spaces,” *The International Journal of Robotics Research*, vol. 29, no. 7, pp. 897–915, 2010.
- [6] S. Karaman and E. Frazzoli, “Sampling-based algorithms for optimal motion planning,” *The International Journal of Robotics Research*, vol. 30, no. 7, pp. 846–894, 2011.
- [7] P. S. Schmitt, W. Neubauer, W. Feiten, K. M. Wurm, G. v. Wichert, and W. Burgard, “Optimal, sampling-based manipulation planning,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2017.
- [8] M. Zucker, N. Ratliff, A. D. Dragan, M. Pivtoraiko, M. Klingensmith, C. M. Dellin, J. A. Bagnell, and S. S. Srinivasa, “CHOMP: Covariant Hamiltonian optimization for motion planning,” *The International Journal of Robotics Research*, vol. 32, no. 9–10, pp. 1164–1193, 2013.
- [9] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel, “Motion planning with sequential convex optimization and convex collision checking,” *The International Journal of Robotics Research*, vol. 33, no. 9, pp. 1251–1270, 2014.
- [10] R. Deits and R. Tedrake, “Computing large convex regions of obstacle-free space through semidefinite programming,” in *Algorithmic Foundations of Robotics XI*. Springer, 2015, pp. 109–124.
- [11] A. Amice, H. Dai, P. Werner, A. Zhang, and R. Tedrake, “Finding and optimizing certified, collision-free regions in configuration space for robot manipulators,” in *Algorithmic Foundations of Robotics XV (WAFR 2022)*. Springer, 2023, pp. 328–348.
- [12] H. Dai, A. Amice, P. Werner, A. Zhang, and R. Tedrake, “Certified polyhedral decompositions of collision-free configuration space,” *The International Journal of Robotics Research*, vol. 43, no. 9, pp. 1322–1341, 2024.
- [13] M. Petersen and R. Tedrake, “Growing convex collision-free regions in configuration space using nonlinear programming,” *arXiv preprint arXiv:2303.14737*, 2023.
- [14] P. Werner, T. Cohn, R. H. Jiang, T. Seyde, M. Simchowitz, R. Tedrake, and D. Rus, “Faster algorithms for growing collision-free convex polytopes in robot configuration space,” in *International Symposium on Robotics Research (ISRR)*, 2024.
- [15] P. Werner, A. Amice, T. Marcucci, D. Rus, and R. Tedrake, “Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2024.
- [16] T. Marcucci, M. Petersen, D. von Wrangel, and R. Tedrake, “Motion planning around obstacles with convex optimization,” *Science Robotics*, vol. 8, no. 84, p. eadf7843, 2023.
- [17] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, “Shortest paths in graphs of convex sets,” *SIAM Journal on Optimization*, vol. 34, no. 1, pp. 507–532, 2024.
- [18] B. P. Graesdal, S. Y. C. Chia, T. Marcucci, S. Morozov, A. Amice, P. A. Parrilo, and R. Tedrake, “Towards tight convex relaxations for contact-rich manipulation,” in *Proc. Robotics: Science and Systems (RSS)*, 2024.
- [19] S. Y. C. Chia, R. H. Jiang, B. P. Graesdal, L. P. Kaelbling, and R. Tedrake, “GCS*: Forward heuristic search on implicit graphs of convex sets,” in *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2024.
- [20] P. E. Hart, N. J. Nilsson, and B. Raphael, “A formal basis for the heuristic determination of minimum cost paths,” *IEEE Transactions on Systems Science and Cybernetics*, vol. 4, no. 2, pp. 100–107, 1968.
- [21] J. López Gómez, “Optimisation-based manipulation planning in convex decompositions of C-free,” MInf Project Report, School of Informatics, University of Edinburgh, 2025.
- [22] M. Toussaint, C. V. Braun, and J. Ortiz-Haro, “NLP sampling: Combining MCMC and NLP methods for diverse constrained sampling,” *arXiv preprint arXiv:2407.03035*, 2024.
- [23] J.-C. Latombe, *Robot Motion Planning*. Kluwer Academic Publishers, 1991.
- [24] P. Tournassoud, T. Lozano-Pérez, and E. Mazer, “Regrasping,” in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 1987, pp. 1924–1928.
- [25] K. Hauser and V. Ng-Thow-Hing, “Randomized multi-modal motion planning for a humanoid robot manipulation task,” *The International Journal of Robotics Research*, vol. 30, no. 6, pp. 678–698, 2011.
- [26] W. Vega-Brown and N. Roy, “Asymptotically optimal planning under piecewise-analytic constraints,” in *Workshop on the Algorithmic Foundations of Robotics (WAFR)*, 2016.
- [27] C. R. Garrett, R. Chitnis, R. Holladay, B. Kim, T. Silver, L. P. Kaelbling, and T. Lozano-Pérez, “Integrated task and motion planning,” *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 4, pp. 265–293, 2021.
- [28] C. R. Garrett, T. Lozano-Pérez, and L. P. Kaelbling, “PDDLStream: Integrating symbolic planners and blackbox samplers via optimistic adaptive planning,” in *Proc. International Conference on Automated Planning and Scheduling (ICAPS)*, vol. 30, 2020, pp. 440–448.
- [29] M. Toussaint, “Logic-geometric programming: An optimization-based approach to combined task and motion planning,” in *Proc. International Joint Conference on Artificial Intelligence (IJCAI)*, 2015, pp. 1930–1936.
- [30] J. Mirabel, S. Tonneau, P. Fernbach, A.-K. Seppälä, M. Campana, N. Mansard, and F. Lamiroux, “HPP: A new software for constrained motion planning,” in *Proc. IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2016, pp. 383–389.

- [31] T. Lozano-Pérez, "Spatial planning: A configuration space approach," *IEEE Transactions on Computers*, vol. C-32, no. 2, pp. 108–120, 1983.
- [32] J. Canny, *The Complexity of Robot Motion Planning*. MIT Press, 1988.
- [33] P. Werner, R. Cheng, T. Stewart, R. Tedrake, and D. Rus, "Superfast configuration-space convex set computation on GPUs for online motion planning," *arXiv preprint arXiv:2504.10783*, 2025.
- [34] T. Schouwenaars, B. De Moor, E. Feron, and J. How, "Mixed integer programming for multi-vehicle path planning," in *Proc. European Control Conference (ECC)*, 2001, pp. 2603–2608.
- [35] R. Deits and R. Tedrake, "Efficient mixed-integer planning for UAVs in cluttered environments," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 42–49.
- [36] —, "Footstep planning on uneven terrain with mixed-integer convex optimization," in *Proc. IEEE-RAS International Conference on Humanoid Robots (Humanoids)*, 2014, pp. 279–286.
- [37] B. Aceituno-Cabezas, C. Mastalli, H. Dai, M. Focchi, A. Radulescu, D. G. Caldwell, J. Cappelletto, J. C. Grieco, G. Fernández-López, and C. Semini, "Simultaneous contact, gait, and motion planning for robust multilegged locomotion via mixed-integer convex optimization," *IEEE Robotics and Automation Letters*, vol. 3, no. 3, pp. 2531–2538, 2018.
- [38] S. Tonneau, D. Song, P. Fernbach, N. Mansard, M. Taïx, and A. Del Prete, "SLIM: Sparse L1-norm minimization for contact planning on uneven terrain," in *Proc. IEEE International Conference on Robotics and Automation (ICRA)*, 2020, pp. 6604–6610.
- [39] T. Cohn, M. Petersen, M. Simchowitz, and R. Tedrake, "Non-Euclidean motion planning with graphs of geodesically-convex sets," in *Proc. Robotics: Science and Systems (RSS)*, 2023.
- [40] V. Kurtz and H. Lin, "Temporal logic motion planning with convex optimization via graphs of convex sets," *IEEE Transactions on Robotics*, vol. 39, no. 5, pp. 3791–3804, 2023.
- [41] R. Natarajan, H. Choset, and M. Likhachev, "Interleaving graph search and trajectory optimization for aggressive quadrotor flight," *IEEE Robotics and Automation Letters*, vol. 6, no. 3, pp. 5357–5364, 2021.
- [42] D. Berenson, S. Srinivasa, and J. Kuffner, "Task space regions: A framework for pose-constrained manipulation planning," *The International Journal of Robotics Research*, vol. 30, no. 12, pp. 1435–1460, 2011.
- [43] L. Jaillet and J. M. Porta, "Path planning under kinematic constraints by rapidly exploring manifolds," *IEEE Transactions on Robotics*, vol. 29, no. 1, pp. 105–117, 2013.
- [44] Z. Kingston, M. Moll, and L. E. Kavraki, "Exploring implicit spaces for constrained sampling-based planning," *The International Journal of Robotics Research*, vol. 38, no. 10–11, pp. 1151–1178, 2019.
- [45] —, "Sampling-based methods for motion planning with constraints," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 1, pp. 159–185, 2018.
- [46] C. W. Wampler and A. J. Sommese, "Numerical algebraic geometry and algebraic kinematics," *Acta Numerica*, vol. 20, pp. 469–567, 2011.
- [47] R. Tedrake and the Drake Development Team, "Drake: Model-based design and verification for robotics," <https://drake.mit.edu>, 2019.
- [48] MOSEK ApS, "The MOSEK optimization toolbox," <https://www.mosek.com>, 2024.
- [49] S. Haddadin, S. Parusel, L. Johansmeier, S. Golz, S. Gabl, F. Walch, M. Sabaghian, C. Jähne, L. Hausperger, and S. Haddadin, "The Franka Emika robot: A reference platform for robotics research and education," *IEEE Robotics & Automation Magazine*, vol. 29, no. 2, pp. 46–64, 2022.
- [50] I. A. Şucan, M. Moll, and L. E. Kavraki, "The Open Motion Planning Library," *IEEE Robotics & Automation Magazine*, vol. 19, no. 4, pp. 72–82, 2012.